# PRATIK SAWANT

## AI ENGINEER | NLP & LLM SYSTEMS | FULL-STACK AI PROJECTS

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## EDUCATION

**B.Tech in CS & Eng (Data Science)**
Vishwakarma Institute of Technology, Pune
Expected Graduation: 2028

## TECHNICAL SKILLS

**Programming**
Python, C++, JavaScript, SQL

**AI / NLP**
PyTorch, TensorFlow, HuggingFace, PEFT/LoRA, LangChain, LangGraph, Prompt Engineering, RAG, Embeddings, Tokenization, NER, Text Classification

**Vector DBs**
Qdrant, ChromaDB, FAISS

**ML Libraries**
Scikit-Learn, Pandas, NumPy

**Backend**
FastAPI, Node.js, React.js

**Tools**
Git, GitHub, Docker, Jupyter Notebook, VS Code

**Cloud**
AWS (Basic), Google Colab GPU

## COURSEWORK

Machine Learning Fundamentals
Data Structures & Algorithms
LangChain & LLM Applications
NLP & Transformer Models
Deep Learning Specialization

## LANGUAGES

English Hindi Marathi

## PROFILE SUMMARY

Aspiring AI Engineer with hands-on experience building production-grade NLP systems, LLM pipelines, and RAG architectures. Proficient in fine-tuning transformer models via PEFT/LoRA, designing vector-database retrieval systems, and developing multi-agent conversational AI. Passionate about advancing the state-of-the-art in language understanding and eager to contribute to enterprise AI teams solving real-world problems.

## PROJECT EXPERIENCE

### Mem-Rooted – Hierarchical Self-Organizing AI Memory System

*Python FastAPI, React, Vector DB ,GPT-4o-mini, NetworkX*

- Designed a novel 4-tier hierarchical memory architecture (ANCHOR→DOMAIN→CLUSTER→INSTANCE) for persona-centric AI, implementing a deterministic Overcome vs Subside cascade using SentenceTransformer embeddings and cosine similarity to auto-place extracted NLP facts without relying on LLM judgment — targeting LoCoMo benchmark F1 >85%.
- Built a 4-way Reciprocal Rank Fusion retrieval engine fusing pgvector semantic search, BM25 keyword, NetworkX graph spreading-activation, and temporal recency scoring; implemented goal-directed exponential decay (spaCy NER + GPT-4o-mini extraction) and an APScheduler Moving Memory engine for autonomous memory consolidation.

https://github.com/Pratiksawant14/Mem-Rooted

### TranslateIQ – Hybrid Enterprise Machine Translation

*Qdrant, PyTorch, PEFT/LoRA, Supabase*

- Engineered an enterprise NLP translation platform with an Adaptive Edge Router that routes segments between OpenRouter cloud LLMs and a locally fine-tuned PyTorch model, cutting token costs by 80%+ while preventing data leakage.
- Implemented JIT domain adaptation via PEFT/LoRA fine-tuning from implicit linguist approval signals; applied RRF fusing Qdrant dense-vector and BM25 sparse search for terminology matching, with a Claude 3.5 Sonnet dual-agent validation layer.

### Intelli-Credit – AI Corporate Credit Appraisal Engine

*LightGBM, SHAP, FastAPI, Supabase*

- Built an 8-node agentic AI pipeline automating corporate loan appraisal — PDF ingestion via Gemini 2.0 Flash OCR, LightGBM credit scoring with SHAP explainability, real-time adverse media crawling, and structured CAM report generation in ~90 seconds.
- Integrated GST fraud reconciliation and a RAG-powered Credit Intelligence Chatbot using PageIndex embeddings and GPT-4o Mini for document-grounded Q&A over financial statements.

### Autonomous CI/CD Healing Agent

*FastAPI, React, LangGraph, GPT-4o, FAISS, Docker*

- Built a self-healing DevOps agent with a LangGraph multi-agent pipeline that clones repos, runs tests in Docker, classifies failure types (syntax/logic/import/type), and auto-generates + commits fixes — zero human intervention required.
- Leveraged FAISS RAG over historical fix embeddings for context-aware code repair; streamed real-time step progress to a React dashboard with automated branch naming.

### Learnify – AI-Powered Learning Platform

*FastAPI, Next.js, OpenAI, ChromaDB, RAG*

- Built an AI platform generating personalized course roadmaps and curating YouTube videos via transcript-based semantic search; engineered a RAG pipeline over ChromaDB embeddings for structured, relevance-ranked learning paths.
- Implemented XP-based gamification with Supabase auth, async API rate-limit handling, and a video intelligence pipeline with transcript extraction and embedding matching.